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**Forests and Human Development: An Analysis of the Socio-Economic
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Tetsuya Michinaka and Motoe Miyamoto

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Forests and Human Development: An Analysis of the Socio-Economic Factors Affecting Global Forest Area Changes

Tetsuya Michinaka ^{*1} and Motoe Miyamoto ^{*2}

ABSTRACT

This study examined the impacts of socio-economic factors on forest area change by human development level of countries. Cluster analysis and panel data analysis were combined to solve the problem of heterogeneity in panel data analysis and the problem of data availability to some extent. The results show that many socio-economic factors have negative impacts on forest area in countries at low levels of human development, but their impacts become positive in countries at higher levels of human development, such as rate of rural population, adult literacy rate, and GDP per capita. The findings of this research remind decision makers to pay attention to increasing countries' level of human development when attempting to prevent deforestation, because people in higher human development index countries tend to be more protective of forests.

Keywords: human development, socio-economic factors, deforestation, cluster analysis, panel data analysis

INTRODUCTION

People treat forests differently at different stages of human development. These differences can be seen throughout thousands of years of human history and in changes in forest area over the past two decades in countries at different stages of development.

Global forest area has been decreasing. Although the pace of this decline has slowed, 83 million ha of forest disappeared from 1990 to 2000, and 52 million ha of forest were eliminated between 2000 and 2010, according to FRA2010 (FAO, 2010). The countries with the largest decreases in forest area in the past two decades include Brazil (55.3 million ha), Indonesia (24.1 million ha), Nigeria (8.2 million ha) and Tanzania (8.1 million ha). Of the 233 countries and regions listed in FRA2010, forest area decreased in 88 countries and increased or remained stable in 136 countries (forest area data are not available for remaining countries). Numerous studies have examined the relationship between forest resources and socio-economic factors, from macro-level analyses utilizing multinational data to micro-level analyses based on field investigations. In particular, poverty (or economic level) and

population growth have been examined as the underlying causes of deforestation (MIYAMOTO, 2010).

Since the late 1980s, the idea that "poverty is the cause and result of environmental degradation" has spread through international organizations (WCED, 1987; WORLD BANK, 1992; JALAL, 1993). As a result, many studies have been conducted to investigate whether poverty accelerates deforestation or not. These studies have identified a number of influential factors and have concluded that the relationship between poverty and the environment is complex (DURAIAPPAH, 1998; DEININGER and MINTEN, 1999; WUNDER, 2001; ANGELSEN and WUNDER, 2003).

Since the 1980s, population growth has been considered one of the most likely causes of deforestation. However, a number of studies have identified diverse results regarding the impact of population growth. ANGELSEN and KAIMOWITZ (1999) reviewed the economic research on deforestation and concluded that although many empirical studies have found a positive correlation between deforestation and population density, the evidence for population growth as a cause of deforestation is weak, because the impact of population might be dependent on other factors, such as infrastructure, soil, and non-agricultural work opportunities. GEIST and LAMBIN (2002) reviewed case studies on deforestation from 150 regions and found many cases in which population factors were not the cause of deforestation. Even in cases in which population was a cause, population growth or population density alone did not cause deforestation.

JHA and BAWA (2006) analyzed the relationship between population growth, the Human Development Index (HDI), and deforestation using statistical data from 30 countries. It showed that 1) when population growth is high and the HDI is low, the deforestation rate is high, and 2) irrespective of

Corresponding author: Tetsuya Michinaka

^{*1} Forestry and Forest Products Research Institute,
Tsukuba 305-8687, Japan

E-mail: zhangyf@affrc.go.jp (T. Michinaka)

^{*2} Hokkaido Research Center, Forestry and Forest
Products Research Institute, Sapporo 062-8516,
Japan

population growth, when the HDI is high, the deforestation rate is low. This research may be the first study to include the HDI in an analysis of the impact of socio-economic factors on forest resources, but models reflecting the relationship between forest and HDI factors are not specified.

The aim of this research is to analyze the impact of socio-economic factors on global forest area changes, from a viewpoint of the relationship between forest and human development level through combining statistical and econometric approaches, cluster analysis and panel data analysis.

Firstly, socio-economic factors include three variables reflecting levels of human development, life expectancy at birth, adult literacy rate and gross domestic product (GDP) per capita, as found in the HDI prepared and issued by the United Nations Development Programme (UNDP) in its Human Development Reports (UNDP, 2011a). UNDP defines human development as “about creating an environment in which people can develop their full potential and lead productive, creative lives in accord with their needs and interests” (UNDP, 2011b). Secondly, because the population size, living area and the extent of development of primary sector (agriculture, forestry, and fishery) are also important, this study also analyzes the impacts of total population, rate of rural population and the gross production value of agriculture. Thus, these six variables cover the level of human development, population size, humans’ geographical relationship with forests, and the development of agricultural production.

The analysis was conducted in two stages. In the first stage, the 205 countries (and regions) were grouped into five clusters using cluster analysis. FRA2010 lists 233 countries, but this study used 205 countries as research objects and excluded the rest due to data availability or small forest area. Our research might deal with the most countries in clarifying the impacts of socio-economic factors. In the second stage, the impacts of the socio-economic factors on changes in forest area were analyzed through a panel data analysis of each of the five clusters. The free software R (R 2.13.1, 2011) and the software TSP (TSP INTERNATIONAL, 2010) were used in the analysis.

As discussed in MICHINAKA *et al.* (2011), by introducing cluster analysis before implementing panel data analysis, the problems of choosing or grouping countries arbitrarily and heterogeneity in panel data analysis can be solved to some extent compared with previous researches; therefore, better results can be obtained.

Research by introducing cluster analysis before panel data analysis has not been found in researches to the impacts of socio-economic factors on deforestation, and some researches are found in other fields, like RICHARDS (2000), MICHINAKA *et al.* (2010 and 2011), VASILESCU *et al.* (2011), ABRAMS *et al.* (2012). RICHARDS (2000) analyzed the household decisions on purchasing apples in six local markets in USA; MICHINAKA *et al.* (2010 and 2011) estimated the demand elasticities for forest products in multiple countries; VASILESCU *et al.* (2011) analyzed the factors affecting earnings for E.U. members; ABRAMS *et al.* (2012) analyzed the price elasticity of demand for water in Sydney.

CLUSTER ANALYSIS

Cluster analysis is described as “the art of finding groups in data” (KAUFMAN and ROUSSEUW, 1990, p. 1) and has been used by biologists and social scientists for over half a century (KAUFMAN and ROUSSEUW, 1987). Cluster analysis is also used in forest science (ATTA-BOATENG and MOSER, 1998; YEHA *et al.*, 2000, MICHINAKA *et al.*, 2010, 2011). For example, ROOS *et al.* (2001) examined differences in Swedish sawmill production strategies using cluster analysis.

There are three reasons for the implementation of cluster analysis prior to the analysis of the impact of socio-economic factors on global forest area changes in this study. Firstly, cluster analysis can produce a general overview that addresses both differences and similarities among countries. That is, by cluster analysis, countries with similarity are grouped together, while countries with less or few similarity are grouped into different groups. Secondly, there are no available and suitable time series data for some countries. It is difficult to undertake an analysis of these countries. Cluster analysis groups countries into clusters based on their similarities across a number of variables, i.e., countries in a cluster based on the cluster analysis have similarity. Therefore, the results of panel data analysis on some countries within a cluster could be extended to all of the member countries in the cluster, which could counter the problem of data availability. Third, some studies have used ad hoc criteria for grouping countries, whereas cluster analysis can identify similarities and group objects scientifically.

Data and their Standardization

Three variables, rate of rural population (%), GDP per capita, and forest change (change in forest area), are used in the cluster analysis to group countries and analyze the impact of socio-economic factors. GDP per capita is one of the most important indices reflecting human development. HDI is not used in this study due to data availability; the number of countries in the UNDP report is much lower than the number of countries whose GDP per capita is known. The rate of rural population is used because we contend that people who live in rural areas and urban areas have different relationships with forests. Data on the rate of rural population (%) and GDP per capita (US\$, purchasing power parity) are for 2008 and sourced from FRA2010. The third variable is forest change. By comparing the forest area data for 1990 and 2010 in FRA2010, it is evident that forest area decreased in some countries, whereas it increased or remained stable in other countries. We allocate the number “1” to those countries with decreased forest area and “0” to the rest. This factor is adopted in the analysis because we contend that countries with similar forest change (decreasing or otherwise) have similarities.

When data are used in the cluster analysis, they are usually standardized. Observations for different variables have different means and variances. If standardization is not performed, large numbers will have more weight in the calculation of distances among objects. In this research, the percentage of the rate of rural population and forest change (1 or 0) is too small compared with the GDP per capita in US

dollars without standardization. Data standardization allows variables to contribute equally. Standardization converts the original data to normalized variables.

The most commonly used standardizing function is z-scores:

$$Z_{ij} = \frac{X_{ij} - \bar{X}_i}{S_i} \quad (1)$$

where X_{ij} is the original data, $\bar{X}_i$ is the mean for the i th variable, and S_i is the standard deviation (SD) for the i th variable. In the partitioning around medoids approach, described in the next section, the mean absolute deviation is used instead of the standard deviation to disperse the impacts of outliers.

Cluster Analysis Approaches and Determination of the Number of Clusters

The partitioning around medoids (PAM), or k-medoids method, a nonhierarchical approach to cluster analysis developed by KAUFMAN and ROUSSEAU (1987), finds k clusters with k representative objects (called medoids, a kind of center for the cluster) in the data and attempts to assign each object in the data set to the proper representative object. Compared

with the k-means method, which aims to partition n objects into k clusters in which each object belongs to the cluster with the nearest mean (centroid), PAM is robust to noise and outliers because the centroid is easily affected by outliers, whereas the medoid is not. Under PAM, a silhouette statistic is calculated that shows how well an object lies within the cluster and it is used to determine the number of clusters. When the silhouette average reaches the maximum, the number of clusters is determined as the best choice. Table 1 shows the results of the cluster analysis. When the number of clusters is five, the silhouette average reaches 0.46, the largest value, as shown in Table 1. Therefore, 205 countries are grouped into five clusters. Usually, the structure of a cluster is said to be stable when the silhouette average is larger than 0.50. Here, 0.46 is also acceptable because it is not far below 0.50. The silhouette averages for the five clusters are 0.62, 0.53, 0.39, 0.42, and 0.34 (Fig. 1). For Clusters 3, 4 and 5, the silhouette averages are not high, meaning that a few countries (2, 3 and 4 countries, respectively) with weak similarities are included in that cluster.

Table 1 Silhouette averages for cluster analysis

Number of Clusters	2	3	4	5	6	7	8	9	Others
Silhouette Averages	0.38	0.42	0.42	0.46	0.44	0.40	0.39	0.39	≤ 0.38

Silhouette plot of pam(x = cluster3v, k = 5, stand = TRUE)

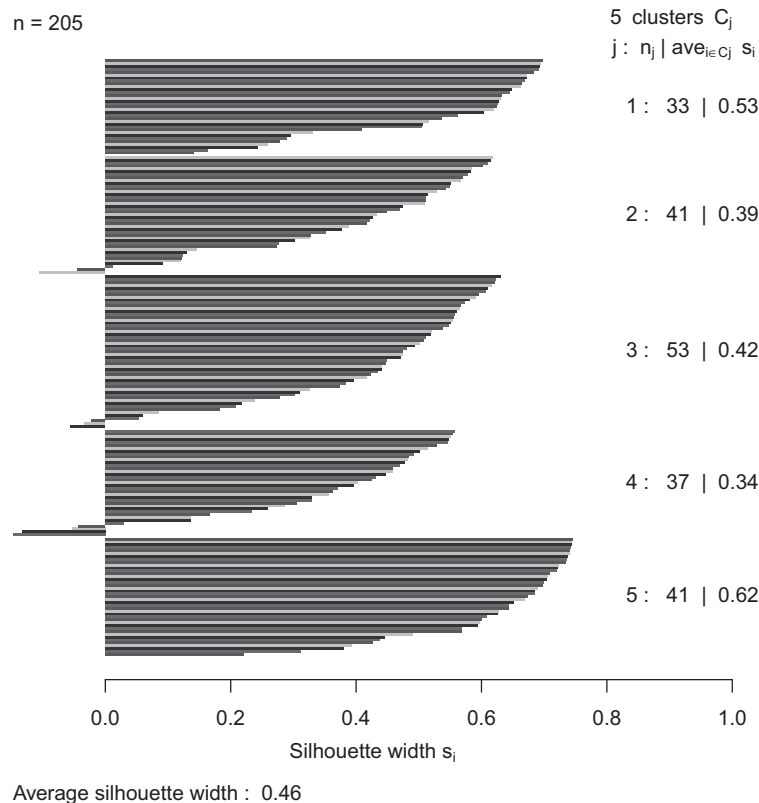


Fig. 1 Silhouette plot for cluster analysis results

Note: the order of clusters shown in the figure is changed in the paper for convenience

Results of Cluster Analysis

By using PAM, 205 countries are grouped into five clusters according to the distances to their medoids, as shown in Fig. 2 and Table 2. Fig. 2 shows the plot of pairs of any two of the three variables. Countries in the same cluster are shown in the same color in Fig. 2. Table 2 shows the detailed results of the cluster analysis. Cluster 1 has the highest mean rate of rural population (72%), the lowest mean GDP per capita (2232 US\$), and decreased forest area. Bangladesh, Cambodia, Lao P.D.R., and Papua New Guinea (PNG) are some examples in this group. Thailand is also included in this cluster due to its high rate of rural population (67%) and decreased forest area. The rate of rural population in Cluster 2 is slightly lower and the GDP per capita is slightly higher compared with Cluster 1, but the forest area did not decrease from 1990 to 2010. China, India, and Viet Nam belong to this cluster. The other three clusters have low rate of rural population. The GDP per capita in Cluster 3 is low, although it is higher than in Clusters 1 and 2. It is also important to note that the countries in Cluster 3, including Brazil, Cameroon, Indonesia, and Malaysia, have decreased forest areas. Clusters 4 and 5 have low rate of rural population and non-decreased forest areas, and the countries in Cluster 5 have the highest GDP per capita. Chile, Gabon, Portugal, and Russian Federation are some of the countries in Cluster 4. Cluster 5 is mainly composed of developed countries, such as Japan, New Zealand, the United Kingdom (UK), and the United States (US). The countries in Clusters 1 and 3 have decreased forest areas, whereas the countries in Clusters 2, 4

and 5 have forest areas that are unchanged or increased, with two exceptions.

Although the first stage of analysis examines 205 countries, due to data availability, only some countries are included in the model specification in panel data analysis in the second stage. The countries that are included in the analysis of the impact of socio-economic factors on forests in the next stage are shown in bold type in Table 2. According to the theory of cluster analysis, countries in the same cluster have similarity. Therefore, the results obtained by the panel data analysis based on some of the countries can represent the entire cluster.

Tests for Cluster Analysis Results

A one-way analysis of means (not assuming equal variances) was tested. This technique compares the means of two or more samples using the F distribution as a variable. The null hypothesis is that the samples in all groups are drawn from the same population. By using R software, the following results are obtained: for GDP per capita, $F = 96.92$, $p\text{-value} < 0.001$ and for rate of rural population, $F = 129.90$, $p\text{-value} < 0.001$. These results show that some clusters have different means—that is, they are not from the same population. For forest change, a one-way analysis of means cannot provide adequate results due to its quantity characteristics. It is evident that some clusters have the same values for forest change, but other clusters have different values. Multiple comparisons between clusters were performed using Tukey's HSD (Tukey's honestly significant difference test, a single-step

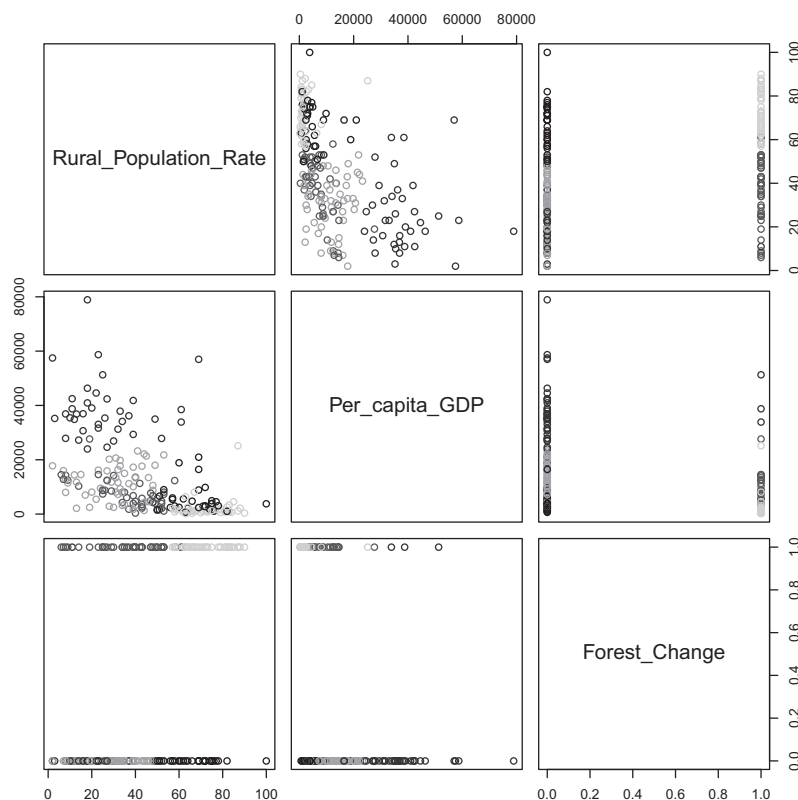


Fig. 2 Pair plot of five clusters for three variables

Table 2 Results of cluster analysis

Clusters	Statistics	Rural Population Rate (%)	Per Capita GDP (US\$)	Forest Change (1 or 0)	Countries in the cluster
(41)	Min	57	200	1	Bangladesh , Benin, Burkina Faso, Burundi , Cambodia, Central African R., Chad, Comoros, D. R. Congo, Eritrea, Ethiopia, Guinea, Guinea-Bissau, Kenya , Lao P. D. R. , Madagascar , Malawi , Mali , Mauritania, Montserrat, <i>Mozambique</i> , Myanmar, Namibia, Nepal, Niger, Pakistan , PNG, Senegal, Sierra Leone, Solomon Islands, Somalia, Sri Lanka, Sudan , Thailand, Timor-Leste, Togo , Trinidad and Tobago , Uganda, Tanzania, Zambia, Zimbabwe
	Mean	72	2232	1	
	Max	90	25173	1	
	SD	10	3985	0	
(33)	Min	48	658	0	Afghanistan, Antigua and Barbuda, Barbados, Bhutan, China , Côte d'Ivoire , Egypt, Fiji , Grenada, Guyana , India , Kiribati, Kyrgyzstan , Lesotho , Maldives, Micronesia, Moldova , Rwanda , Saint Helena, Ascension and Tristan da Cunha, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Samoa, Swaziland , Tajikistan , Tonga, Turkmenistan, Tuvalu, Uzbekistan, Vanuatu, Viet Nam , Wallis and Futuna Islands, Yemen
	Mean	67	5205	0	
	Max	100	20970	0	
	SD	11	4948	0	
(41)	Min	6	388	1	Albania , Algeria, Angola, Argentina , Armenia , Belize, Bolivia, Bosnia and Herzegovina, Botswana , Brazil , Cameroon , Colombia , Congo , D. P. R. Korea, Dominica, Ecuador , El Salvador , Georgia, Ghana , Guatemala , Haiti, Honduras , Indonesia , Jamaica , Kazakhstan , Liberia, Malaysia , Mauritius , Mexico , Mongolia , Nicaragua , Nigeria, Niue, Northern Mariana, Panama , Paraguay , Peru , S. Korea , Suriname, Virgin, Venezuela
	Mean	37	7813	1	
	Max	58	27658	1	
	SD	14	5229	0	
(53)	Min	2	1363	0	Samoa, Aruba, Azerbaijan, Belarus, Bulgaria , Cape Verde, Chile , Cook, Costa Rica , Croatia , Cuba, Djibouti, Dominican R. , Estonia , F. Polynesia, Gabon , Gambia , Hungary , Iran , Iraq, Jordan, Latvia , Lebanon, Libyan, Lithuania , Marshall, Montenegro, Morocco , Antilles, New Caledonia, Palestinian, Oman, Palau, Philippines , Poland , Portugal , Puerto Rico, Romania , Russia , Pierre and M., Sao T. P., Serbia, Seychelles, Slovakia , S. Africa , Syrian , Yugoslav, Tunisia , Turkey , Turks and Caicos, Ukraine , Uruguay , Western Sahara
	Mean	31	11522	0	
	Max	53	23254	0	
	SD	13	6163	0	
(37)	Min	2	23991	0	Andorra, Australia , Austria , Bahamas, Bahrain, Belgium , British Virgin, Brunei Darussalam , Canada , Cyprus , Czech , Denmark , Equatorial Guinea, Finland , France , Germany , Greece , Iceland, Ireland , Isle of Man, Israel , Italy , Japan , Jersey, Kuwait, Luxembourg, Netherlands , New Zealand , Norway , Saudi Arabia, Slovenia , Spain , Sweden , Switzerland , UAE, UK, USA
	Mean	26	38431	0	
	Max	69	78922	1	
	SD	16	10963	0	
All	Min	2	200	0	205 countries
	Median	43	7716	0	
	Mean	45	12762	0.4	
	Max	100	78922	1	
	SD	23	14127	0.5	

Note: 1. Some names are shortened for convenience.

2. Numbers in brackets are the number of countries included in the cluster.

3. Countries in **bold** type are included in the panel data analysis in the next step.

4. For forest change, "1" implies a decrease in forest area and "0" implies otherwise.

multiple comparison procedure) with multiple comparisons of means at a 95% family-wise confidence level. The results show that the differences in some pairs are not significant on an individual variable basis. By determining whether a pair with no significant difference appears simultaneously in three dimensions of variables, it can be concluded that a difference in any two clusters is significant in at least one dimension. For example, compared with Cluster 1, Cluster 2 has a lower but similar rate of rural population, a higher but similar GDP per capita, and an opposite forest change. Therefore, the difference between Cluster 1 and Cluster 2, at least for one of the three variables, is verified by multiple comparisons.

IMPACTS OF SOCIO-ECONOMIC FACTORS ON FOREST AREA

Models and Data

"A longitudinal, or panel data set is one that follows a given sample of individuals over time, and thus provides multiple observations on each individual in the sample" (HSIAO, 2003). Panel data analysis has the following advantages: (a)

it can solve problems that cannot be solved by either time-series data or cross-sectional data alone; (b) it can increase the degrees of freedom when using a large number of data points; and (c) it can improve the efficiency of estimations (HSIAO, 2003). Panel data analysis is a useful tool, especially when the sample is short and wide, such as when the sample contains a large number of cross-sectional individuals or units over a small number of time periods. The model for panel data analysis takes the following form:

$$y_{it} = \alpha_i + \beta'X_{it} + u_{it}, \quad i = 1, \dots, N \text{ and } t = 1, \dots, T \quad (2)$$

Usually, there are three basic kinds of econometric models by panel data analysis: the pooled OLS model, the fixed effects (FE) model, and the random effects (RE) model. For pooled OLS models, the intercept and slopes are same for all the individuals, without considering the individual effects. For FE models, the intercepts α_i are fixed and different for different individuals, but the slopes (coefficients) remain the same for all individuals, treating the effects α_i as fixed. RE models are similar to FE models except treating the effects as

unobserved random variables by a generalized least squares procedure. For the above model, in this research, the function uses the natural logarithm and takes the following empirical form:

$$\ln(y_{it}) = \beta_0 + \beta_1 \ln(LE_{it}) + \beta_2 \ln(ALR_{it}) + \beta_3 \ln(GDP_{it}) + \beta_4 \ln(POP_{it}) + \beta_5 \ln(RRP_{it}) + \beta_6 \ln(GAP_{it}) + e_{it} \quad (3)$$

where y_{it} stands for the forest area in country $i = 1, \dots, N$ during year $t = 1, \dots, T$, LE is life expectancy at birth (years), ALR is the adult literacy rate (%), GDP is GDP per capita, POP is the total population (persons), RRP is the rate of rural population (%), GAP is the gross production value of agriculture, and e_{it} is the error term. For a logarithm function, the slope coefficients represent the rate of change, or elasticity. For example, β_1 indicates that when LE changes 1%, the forest area will change β_1 percent.

FRA2010 provides data for forest area in 1990, 2000, 2005 and 2010. Because we need data for the same time interval to conduct a panel data analysis, forest area data for 1995 are interpolated by obtaining the average of the data in 1990 and 2000. Life expectancy at birth (years), adult literacy rate (%), and GDP per capita (constant 2008 PPP US\$) are sourced from UNDP. The gross agricultural product (constant 2004-2006 US\$), total population and rural population data are from FAO (2011).

Results

In this study, some tests are implemented in order to get unbiased results, shown as in Table 3. (1) F test is implemented for panel data poolability, individual effect, to compare FE models versus pooled OLS models. The F test results show that FE model estimations are better than pooled

OLS models in all clusters. (2) Breusch-Pagan Lagrange Multipliers are calculated to compare RE models with pooled OLS models (Breusch and Pagan, 1980). It is found that all the RE models are superior to pooled OLS models. (3) Hausman test is implemented to compare FE and RE models. The results show that FE models are better than RE models in all clusters except Cluster 1. (4) Testing for serial correlation is undertaken by using Breusch-Godfrey/Woodridge test (BALTAGI and LI, 1995, WOODRIDGE, 2002, CROISSANT and MILLO, 2008). It shows that all the clusters have serial correlation in idiosyncratic errors. (5) Testing for heteroskedasticity is undertaken by studentized Breusch-Pagan test (BREUSCH and PAGAN, 1979). It is found that all the clusters have the problem of heteroskedasticity at the different levels of significance, 1%, 5%, or 10%, except Cluster 2, which is not heteroskedastic. (6) Testing for cross-sectional dependence, or contemporaneous correlation, is undertaken by using Pesaran CD test (PESARAN, 2004). It is found that there is no cross-sectional dependence in Clusters 1 through 4, with an exception for Cluster 5.

For Clusters 1, 3, and 4, to deal with the problems of both serial correlation and heteroskedasticity, general feasible generalized least squares models are estimated, as estimations of general feasible generalized least squares are “robust against any type of intragroup heteroskedasticity and serial correlation” (CROISSANT and MILLO, 2008). For Cluster 2, serial correlation exists, but not heteroskedastic and no cross-sectional dependence; AR(1) is implemented in order to correct serial correlation errors. For Cluster 5, robust covariance matrix of parameters was estimated for dealing with cross-sectional dependence and serial correlation by using Pesaran CD approach (BECK and KATZ, 1995).

Panel data analysis was applied to every cluster, and the results are shown in Table 4. We will now discuss which

Table 3 Test results for 5 clusters

Items	1	2	3	4	5	Remarks
(1) Testing for poolability (comparing FE models vs pooled models, F test)	p-value < 0.001	p-value < 0.001	p-value < 0.001	p-value < 0.001	p-value < 0.001	FE models are better than pooled models for all clusters
(2) Testing for poolability (Breusch-Pagan Lagrange Multiplier for RE models)	p-value < 0.001	p-value = 0.002	p-value < 0.001	p-value < 0.001	p-value < 0.001	RE models are better than pooled model for all clusters
(3) Comparing models: RE vs FE (Hausman test)	p-value = 0.967	p-value = 0.008	p-value < 0.001	p-value < 0.001	p-value = 0.020	FE models are better than RE in 1 to 4
(4) Testing for serial correlation (Breusch-Godfrey/Wooldridge test)	p-value = 0.020	p-value < 0.001	p-value < 0.001	p-value < 0.001	p-value < 0.001	Serial correlation exist in all clusters
(5) Testing for heteroskedasticity (Breusch-Pagan test)	p-value = 0.034	p-value = 0.411	p-value = 0.062	p-value = 0.025	p-value = 0.001	Clusters 1, 3, 4 and 5 are heteroskedastic; cluster 2 is not heteroskedastic
(6) testing for cross-sectional dependence (Pesaran CD test)	p-value = 0.752	p-value = 0.286	p-value = 0.220	p-value = 0.861	p-value = 0.007	No cross-sectional dependence except in Cluster 5

Note: for (6), p-values from FE models are shown here, even though RE models have similar results.

Table 4 Results for panel data analysis for 5 clusters

Variables	1	2	3	4	5
Life expectancy at birth (years)	0.01	0.19	-0.31***	0.08	0.50
Adult literacy rate (%)	-0.72***	-0.21	-0.41***	0.13	1.60***
GDP per capita	0.09	0.05	-0.04*	0.05***	0.23**
Total population	0.27	0.03	-0.19***	-0.12	-0.27
Rural population rate (%)	0.75	-0.45***	-0.02	-0.16**	0.08**
Gross production value of agriculture (US\$)	0.01	0.15**	-0.01	0.01	0.00

Note: ***: significant at 1% level by t values; **: significant at 5% level; *: significant at 10% level.

factors affect forest area changes, significantly or otherwise.

In Cluster 1, only adult literacy rate is significant; it has a negative impact on forest area. This means that as people obtain more education and literacy, they tend to cause forest area to decrease in the countries in Cluster 1. Other factors, including GDP per capita and population, do not affect forest area significantly, which may be due to the data quality in countries with high rate of rural population and low per capita income.

In the countries in Cluster 2, the rate of rural population and gross production value of agriculture are significant, while other factors are not. The rate of rural population has a negative impact on forest area, but the increase in gross production value of agriculture can cause forest area to increase. The rate of the rural population shows a decreasing trend globally, from 57% in 1990 to 49% in 2010. For most countries (with some exceptions, such as Sri Lanka and Zambia), the rate of rural population has decreased due to urbanization. Although the rate of rural population has a negative impact on changes in forest area, the pressure of the increase in rate of rural population on forests decreases. As for the second significant factor, agriculture includes forestry, hunting, fishing, the cultivation of crops and livestock production (WORLD BANK, 2009). The countries in this cluster, such as Viet Nam and China, have increased forest areas, possibly because of the important role of the development of agriculture, although the rural population counteracts this increase in forest area.

In Cluster 3, all the countries have decreased forest areas; all factors except the rate of rural population and agricultural gross production value had negative impacts on forest area. It seems that people tend to cause forest area to decrease when their income increases and their life situations improve.

In Cluster 4, the rate of rural population have negative impacts on forest area, while GDP per capita has a positive impact on forest area, but other factors do not have significant impacts. The countries in Cluster 4 have lower rate of rural population and higher GDP per capita than those in Clusters 1, 2 and 3. Although the rural population imposes pressure on the increase in forest area, the decreases in the rate of rural population alleviate this pressure.

In Cluster 5, the adult literacy rate, GDP per capita, and rate of rural population are significant and have positive impacts on forest area. This cluster is composed of the highest human development countries with the lowest rate of rural population and the highest GDP per capita. It shows that the

increase in income can further contribute to the afforestation in the countries in Cluster 5. Positive impacts from the rate of rural population imply that the rural population is important to maintain or increase forest area. In other words, the decreased rate of rural population places pressure on maintaining the forest area in countries at high level of human development.

Now let us check the results in the light of factors. Life expectancy at birth and total population have significant and negative impacts in Cluster 3 but not significant in other clusters. The adult literacy rate has negative impacts in Clusters 1 and 3, both of which are at lower levels of human development, but it has a positive and significant impact in Cluster 5, which has the highest level of human development. The elasticities of adult literacy rate have also decreased from 0.72 in Cluster 1 to 0.41 in Cluster 3. GDP per capita is significant in the last three clusters, but their sign are different, changing from negative to positive, as their levels of human development increase. Rural population has a significant and negative impact in Clusters 2 and 4, and a significant and positive impact in Cluster 5, but not significant in Clusters 1 and 3. The elasticity for rate of rural population has decreased from 0.45 in Cluster 2 to 0.16 in Cluster 4 and 0.08 in Cluster 5. Gross production value of agriculture is significant only in Cluster 2. Cluster 2 is composed of countries with high rate of rural population, but an increasing forest area. It seems that the development of agriculture helps afforestation and reforestation in these countries. These impacts indicate that the impact of some factors change from negative to positive, such as adult literacy rate, GDP per capita and rate of rural population, as the level of human development increases; some other factors changed their significant negative signs to not significant, such as life expectancy at birth, total population.

CONCLUSIONS AND DISCUSSION

This research combines cluster analysis and panel data analysis. Five clusters were obtained by cluster analysis. Through cluster analysis, countries with similarity are grouped into one cluster, while countries with less similarity are gathered in different clusters. Panel data analysis is implemented to countries in each cluster, and better results were obtained because the heterogeneity issue in panel data analysis is improved as member countries in each cluster have similarity (MICHINAKA *et al.*, 2011). In a cluster, for some countries, suitable data for econometrics analysis are available,

while for other countries there are no suitable data. Results by panel data analysis to those countries in which data are available could represent the whole cluster and extend the results to all the countries in the same cluster due to existence of similarity among countries in the same cluster according to the cluster analysis theory. In this way, the problem of data availability is solved, to some extent.

JHA and BAWA (2006) associates forest area changes with the HDI index and calculates the correlation coefficients, paving the way for this quantitative analysis of their impacts. Our study may be the first to analyze the impacts of socio-economic factors, including three HDI variables, using econometric analysis, especially combining cluster analysis with panel data analysis.

This study found that many factors have negative impacts on forest area in countries at low levels of human development, but that the impacts become positive or not significant in countries at higher levels of human development. This finding implies that the relationship between forests and people changes when the level of human development changes.

The rate of rural population, rather than agricultural population or the rate of agricultural population, is used as a factor in this analysis. When people live in rural areas, it is easier for them to access forests. This close relationship with forests is reflected in the construction of homes, the use of fuel wood and timber in daily life, and the extension of their farmland or the development of new farmland by converting forests to agriculture fields, which may cause forest degradation and deforestation. The rate of rural population not only reflects changes in the rural population and the share of the rural population in the total population, but it also reflects urbanization and the development of human society. The differences between the rate of rural population and the rate of agricultural population might be caused by the speed of urbanization and the development of non-agricultural industries, such as the manufacturing and service industries, as well as the development of agriculture. Both seem to be possible causes of deforestation. It is difficult to determine their priority due to the data quality, especially for developing countries.

Total population is used as a factor in this research, but population density is not. For a country, population density is the total population divided by its land area; the land area is stable during the entire period of analysis, with some exceptions. The logarithm form is used in the estimation of coefficients, and the coefficients of factors reflect rates of change related to these factors and forest area. The coefficients of factors do not change if land areas are added to the total population (i.e., if population density is used), although this process will change the intercepts.

One of the most frequently discussed theories in recent years concerning the relationship between environment and economic development is the "Environmental Kuznets Curve" (EKC), which presents the relationship between economic level and environmental degradation as an inverted-U shape. This theory states that as the economic level increases, environmental degradation also increases; once a certain level is reached, however, environmental degradation decreases.

GDP, GNI and income are used as indicators of economic level. According to Stern's review (2004) of studies on the EKC, there have been many attempts to verify the EKC, but empirical evidence is weak, and a consensus has not been reached. In this research, the change in elasticities in GDP per capita, from negative 0.04 to positive 0.05 and 0.23, as the levels of human development rise, implies the course that income is a negative factor in some countries with low levels of human development, whereas in countries at a high level of human development, income changes to a positive factor. This finding seems to be in accordance with the Environmental Kuznets Curve and the U hypothesis of forest resources (STERN, 2004; NAGATA *et al.*, 1993, *etc.*).

Previous researches have also dealt with the deforestation version of the EKC. One affirmative report from BHATTARAI and HAMMIG (2001) analyzed the relationship between deforestation and income for 66 tropical countries from 1972 to 1991 and found strong evidence in support of the EKC in Latin America, Africa and Asia. In contrast, BARBIER and BURGESS (2001) presented a critical report using panel data for tropical countries from 1961 to 1994 and found that the relationship between agricultural land expansion and income varied by region. These authors concluded that this relationship is not necessarily an inverted-U curve. Van and AZOMAHOU (2007) analyzed panel data for 59 developing countries from 1972 to 1994, but their study failed to find evidence supporting the EKC in relation to deforestation. Another way of discussing the deforestation version of the EKC is "Forest Transition", a theory by MATHER (1990, 1992) of the historical development of forest resources based on the cases of developed countries. Mather hypothesized that as economies develop, forests initially decline but eventually begin to increase. The debate on this concept continues to this day (NAGATA *et al.*, 1993; MATHER *et al.*, 1999; RUDEL *et al.*, 2005).

Life expectancy at birth and adult literacy rate have negative impacts to countries at low level of human development, but not significant in high level countries. It seems that these two factors in the human development also follow the similar course with income in the light of the relationship with forest. If this is true, it is not enough to link only income with environment as in EKC or the U hypothesis of forest resources without adding life expectancy and an education factor as reflected in HDI.

The findings of this research can remind decision-makers to pay attention to increasing countries' level of human development when attempting to prevent deforestation in countries at low level of human development, i.e., forest policy should be considered with policy on human development comprehensively. Policies that attempt to increase the levels of HDI in these countries are needed because people in higher HDI countries tend to be more protective of forests. The introduction of the HDI index is also a reminder of the importance of life expectancy and education in addition to income level.

This paper introduced cluster analysis before panel data analysis for the first time in this field, by which the problem of heterogeneity in panel data analysis is improved, and tried to avoid biased estimations by solving the problems of serial correlation, heteroskedasticity, and cross-sectional

dependence. Maybe due to the reliability of data, some of the coefficients are not significant as expected, such as the rate of rural population in Clusters 1 and 3, which are consist of countries with deforestation. As the data availability and quality get improved in the future, the results from cluster analysis and panel data analysis might be expected to get improved too.

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